

LiteFusion-DETR: Lightweight Dual-Branch DETR for Efficient Multi-Modal UAV Detection

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Keywords: UAV Detection · Multimodal Fusion · Transformer · Visible-Infrared

The Rise of the Low-Altitude Economy

Unmanned aerial vehicles (UAVs) power a fast-growing economy spanning **all three industrial sectors**.



UAVs enabling diverse
low-altitude operations



Power-line Inspection



Aerial Photography



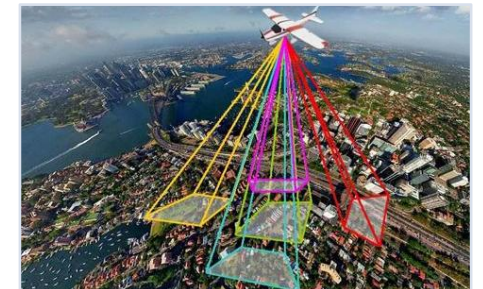
Airborne Law Enforcement



Emergency Rescue



Logistics Network



Urban Mapping

Driving the economy · strengthening public services · serving national defense

But Drone Misuse Threatens Security

Frequent **illegal / reckless drone flights** endanger both military and civilian safety — and hold back the low-altitude economy.

Military Conflict



Apr 2024 — Iran–Israel escalation:
drone strikes trigger a global
political shockwave

Civil-Aviation Disruption



Sep 2024 — Tianjin Airport:
two days of rogue-drone interference,
mass flight delays / cancellations

Critical-Site Intrusion



Jun 2025 — Ukraine strike on
Russian strategic bomber base,
a severe national-security threat

Why Multi-Modal Anti-Drone Detection?

RGB Camera

Strengths — rich texture & color information.

Weaknesses — fails under low light (night, dusk); degraded by glare / backlight / flash; vulnerable to camouflage & background clutter.

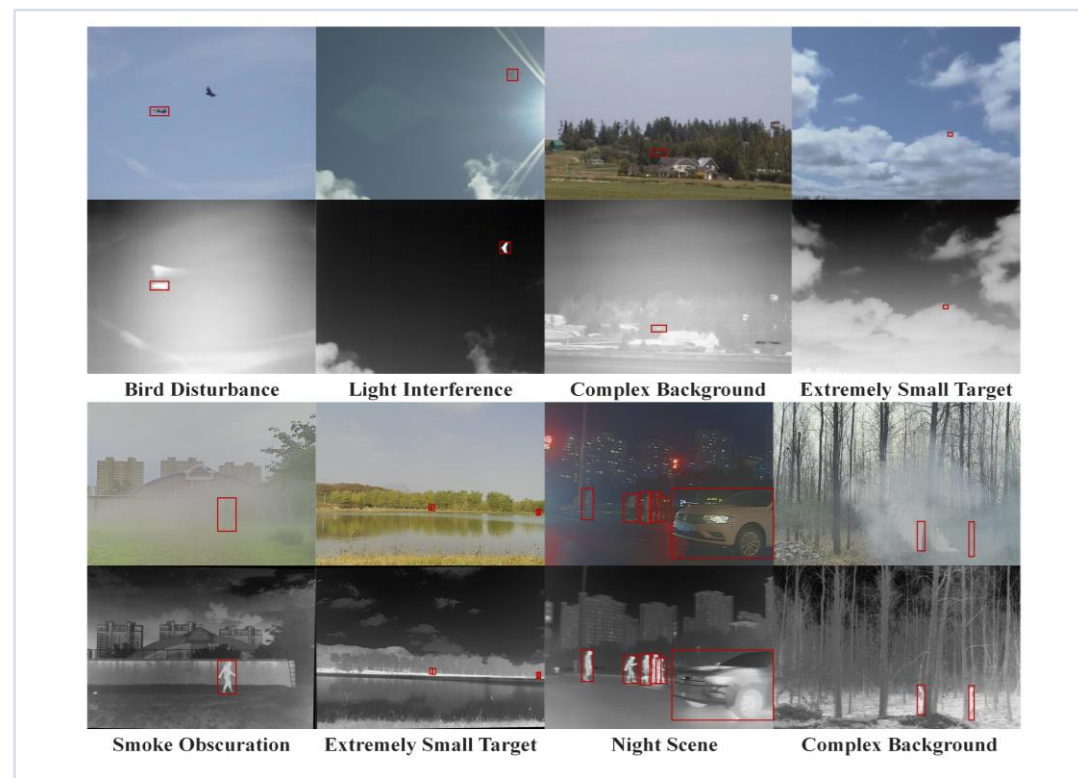
Infrared (IR) Sensor

Strengths — invariant to illumination; captures thermal radiation (works at night).

Weaknesses — low spatial resolution; background thermal noise (buildings, birds); hard to distinguish similar-temperature objects.

⇒ **No single modality is robust enough — fusing RGB + IR is the key.**

Detecting **Low-Slow-Small (LSS) drones** is hard — and no single sensor is robust enough.



Sample RGB-IR pairs from CUAS-EOIR & M3FD: tiny targets, glare, clutter, bird interference.

Mainstream RGB–IR Fusion Detection — and Where It Falls Short

Category	Representative Work	Fusion Strategy	Limitation
Registration + fusion	SuperFusion (Tang '22)	Explicit registration + attention fusion	Registration errors cascade downstream ; fragile in dynamic UAV scenes; 47.65M params
Two-stream CNN fusion	SLBAF-Net (Cheng '23)	Lightweight YOLOv5-based fusion	Weak inter-modal interaction — cannot suppress misalignment interference
Two-stream backbone fusion	TwoStream-YOLOv8 (Mu '24)	Progressive YOLOv8 backbone fusion	Modality noise interferes mutually; low fusion efficiency
Transformer fusion	Fusion-DETR (Huang '25)	Cross-modal transformer encoder	Redundant self-attention — 79.7M params ; degrades sharply under weak alignment
Dynamic modality selection	RTM-UAVDet (Wang '24)	Single-modality dynamic switching	Discards complementary cues — pseudo multi-modal
Data-level fusion	SuperYOLO (Zhang '23)	Pixel-wise addition	Highly sensitive to modality noise

Our design: a lightweight dual-stream end-to-end framework — **position-aware gated cross-attention** for robust semantics, **channel–spatial attention distillation** for small-target cues.

Three Core Challenges of RGB–IR Fusion

Three concrete bottlenecks shape our design — each motivates one component of LiteFusion-DETR.

① Weak-Alignment Semantic Distortion

Platform vibration, parallax and different imaging physics misalign RGB–IR pairs.

67% of DroneVehicle pairs are offset ≥ 8 px.

Naive concatenation → “ghosting” & semantic conflict.

↓ *addressed by*

PCMGT

position-aware gated cross-attention

② Low-Level Redundancy & Small-Target Loss

Early layers (S3 / S4) are unfiltered — IR thermal noise mixes with RGB texture, drowning tiny UAV cues.

Prior fusion targets image generation or deep features — not multi-scale small-target detection.

↓ *addressed by*

DBAF

channel-wise attention distillation

③ End-to-End Efficiency Bottleneck

Applying end-to-end detectors to the multi-modal task inflates parameters & FLOPs.

Fusion-DETR reaches 79.7M params with poor inference speed — accurate but too heavy to deploy.

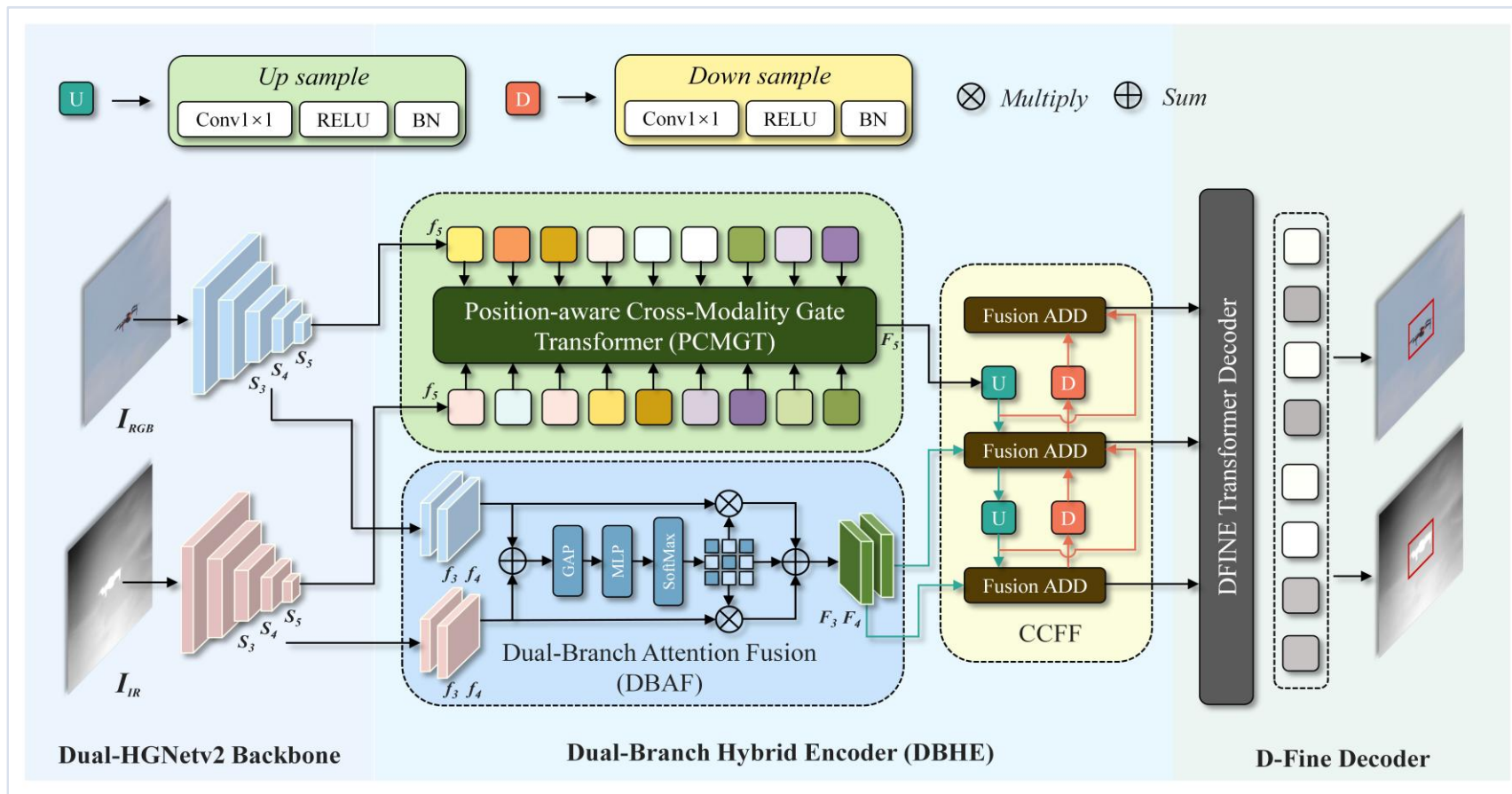
↓ *addressed by*

Lightweight Design

dual-branch hybrid encoder, no self-attention

LiteFusion-DETR Architecture

A single end-to-end pipeline — cross-modal alignment, fusion and detection are jointly optimized.



RGB-IR pair $\rightarrow$ dual HGNetv2 (S_3 - S_5) $\rightarrow$ DBHE (PCMGT + DBAF) $\rightarrow$ CCFF $\rightarrow$ D-FINE decoder $\rightarrow$ detections.

① Dual-HGNetv2 Backbone

Parallel RGB / IR extraction $\rightarrow$ multi-scale features S_3, S_4, S_5 .

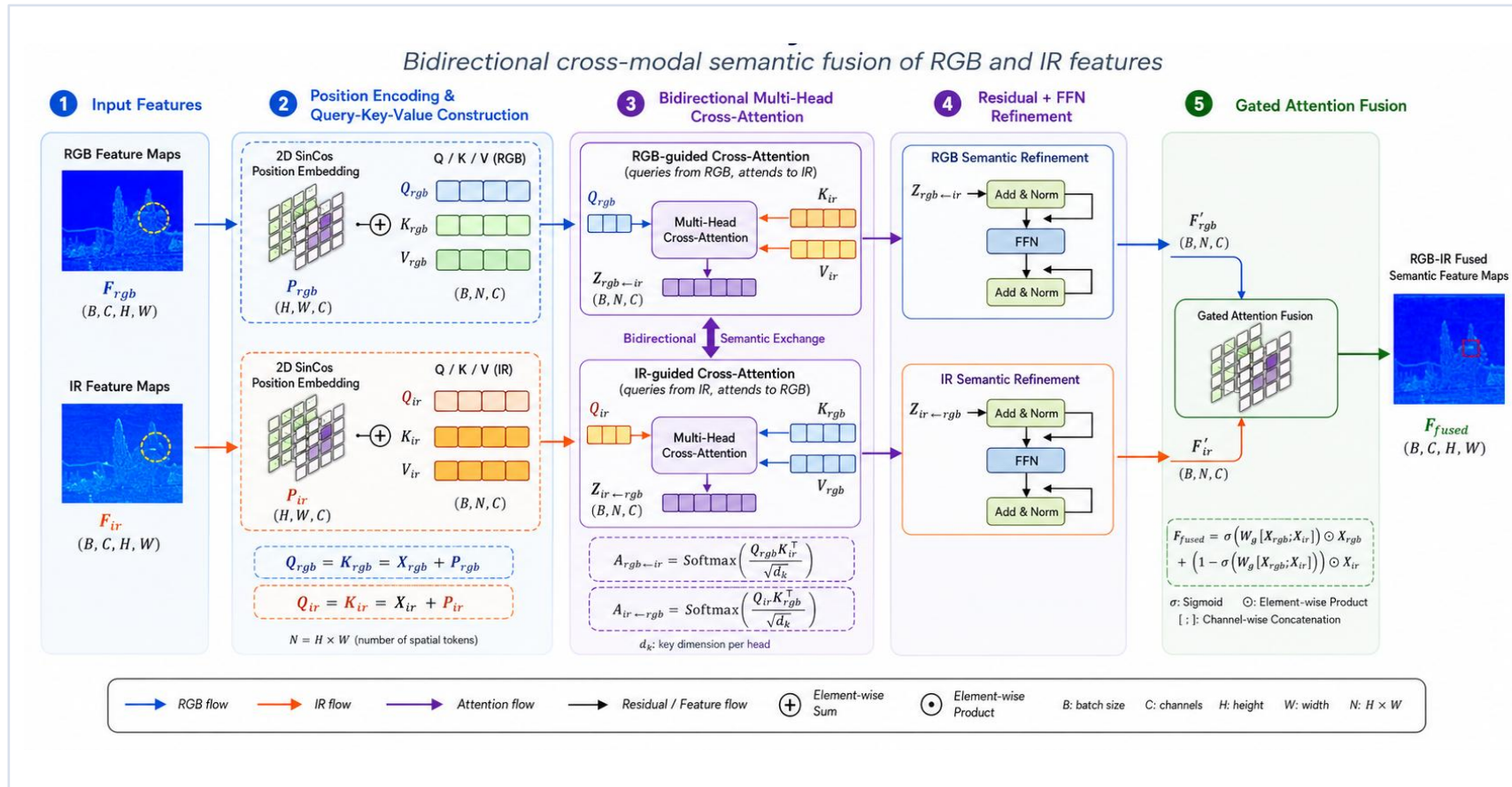
② Dual-Branch Hybrid Encoder

Dual-granularity fusion: **PCMGT** on high-level S_5 , **DBAF** on low-level S_3 / S_4 .

③ D-FINE Decoder

End-to-end set prediction $\rightarrow$ final detections; no NMS, no separate registration stage.

PCMGT: Position-aware Cross-Modality Gate Transformer



$$\text{Gated fusion: } F_{fused} = \sigma(W_g[X_{rgb}; X_{ir}]) \odot X_{rgb} + (1 - \sigma(W_g[X_{rgb}; X_{ir}])) \odot X_{ir}$$

High-Level Semantic Fusion (S5)

① Position-aware

2D SinCos encodings with modality-specific temperatures → robust to RGB-IR misalignment.

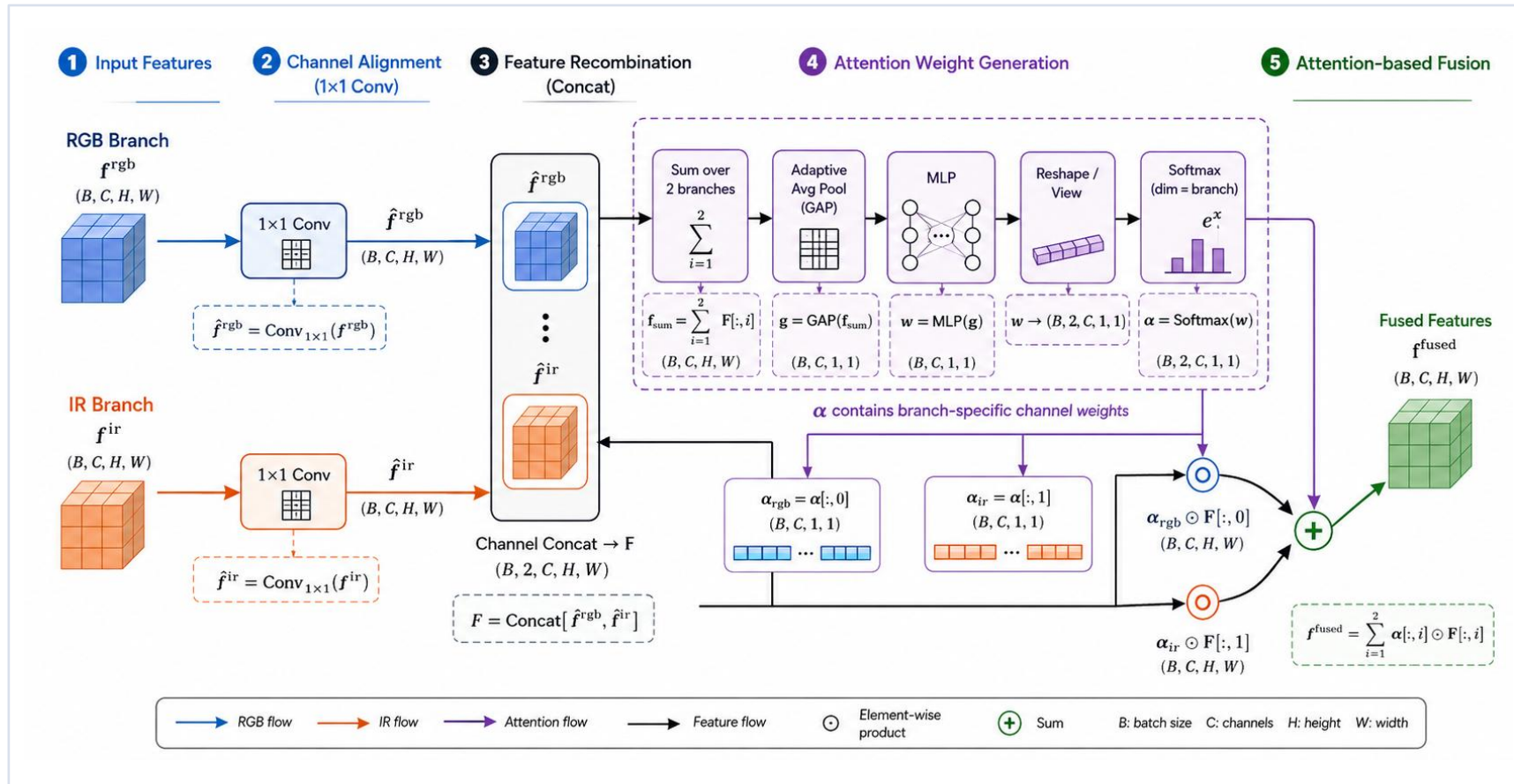
② Bidirectional cross-attention

IR queries pull RGB texture; RGB queries pull IR thermal cues. No self-attention → minimal parameter overhead.

③ Gated fusion

Content-aware sigmoid gate keeps reliable cues, suppresses misalignment “ghosting”.

DBAF: Dual-Branch Attention Fusion



$$\text{Fused feature: } f^{fused} = \sum_{i=1}^2 \alpha_i \odot f_i \quad \alpha_i: \text{softmax attention weights over the RGB and IR branches}$$

Low-Level Feature Distillation (f3, f4)

① Why low-level?

Fine spatial details are vital to localize Low-Slow-Small drones — but raw RGB-IR pairs are redundant and costly.

② Channel-wise attention

Sum → GAP → MLP → Softmax across the two branches → adaptive per-channel fusion weights.

③ Compress & enhance

Distills redundant pairs into one compact map: -0.26M params, -1.73G FLOPs, +1.36 mAP.

Experimental Setup

Three widely used visible–infrared benchmarks + a unified training protocol.

M3FD

General multimodal benchmark

- 4,200 aligned RGB–IR pairs
- 6 classes: person, car, bus, motorcycle, truck, lamp
- Diverse scenes: day / night, smoke, clutter

DroneVehicle

Aerial, weakly-aligned

- 28,439 RGB–IR pairs · 17,990 train / 1,469 test
- 5 classes: car, truck, bus, van, freight car
- Real spatial misalignment; IR annotations as ground truth

CUAS-EOIR

Anti-drone scenario

- 5,977 images @ 640×512 · 4,604 / 616 / 614 split
- Small-UAV detection; ~20% synthetic
- Spatially & temporally aligned EO–IR

Implementation Details

Backbone

Dual HGNetv2

Hardware

RTX 3090 · CUDA 12.1

Decoder

D-FINE

Input

640 × 640

Optimizer

AdamW + EMA (0.999)

Metrics

COCO mAP · AP50

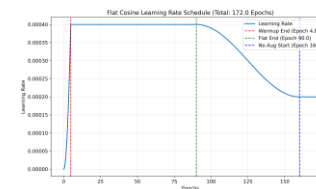
Epochs

172 · batch 8

Speed

FPS @ batch = 1

Flat Cosine LR Schedule (172 epochs)



Comparison with State of the Art

M3FD general fusion benchmark

Method	mAP50	mAP
D-FINE-IR	85.4	54.6
SuperFusion	83.5	56.0
Fusion-Mamba	88.0	61.9
Fusion-DETR	90.8	62.3
baseline-RGB	86.4	55.7
baseline-IR	87.4	56.5
LiteFusion-DETR	92.5	63.26

DroneVehicle weakly-aligned aerial

Method	mAP50	Params	FPS
MFPT	82.0	47.7M	51.2
CMADet	82.0	33.3M	208.3
COMO	85.3	16.3M	135.1
Fusion-DETR	81.3	79.7M	—
baseline-IR	79.8	10.2M	149.6
LiteFusion-DETR	83.6	12.6M	181.3

CUAS-EOIR anti-drone

Method	mAP	mAP50
RT-DETRv2-IR	67.2	91.0
D-FINE-IR	69.3	92.4
TwoStream-YOLOv8	74.3	94.6
YOLOv11-RGBT	71.2	92.6
Fusion-DETR	65.9	91.7
baseline-IR	70.3	92.7
LiteFusion-DETR	78.2	97.7

New SOTA on **M3FD** (+1.7 mAP50 over Fusion-DETR) and **CUAS-EOIR** (+3.1 mAP50 over TwoStream-YOLOv8).

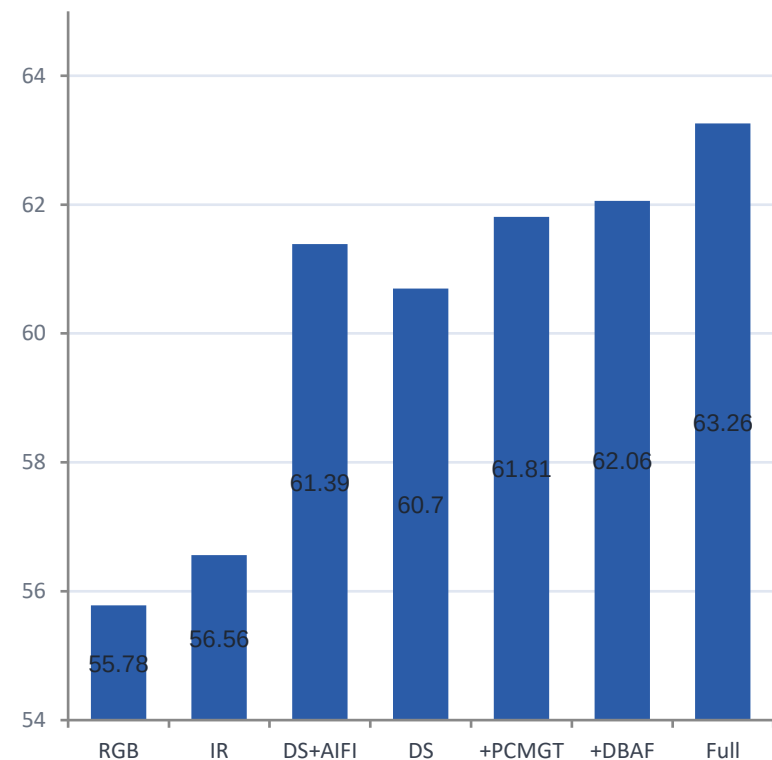
Only **12.64M parameters** — 6.3× fewer than Fusion-DETR (79.7M) — with real-time inference on a single RTX 3090.

Ours in red. Single-modality baselines share the same HGNetv2 backbone and training protocol. Full per-class and efficiency results in the paper.

Ablation Study on M3FD

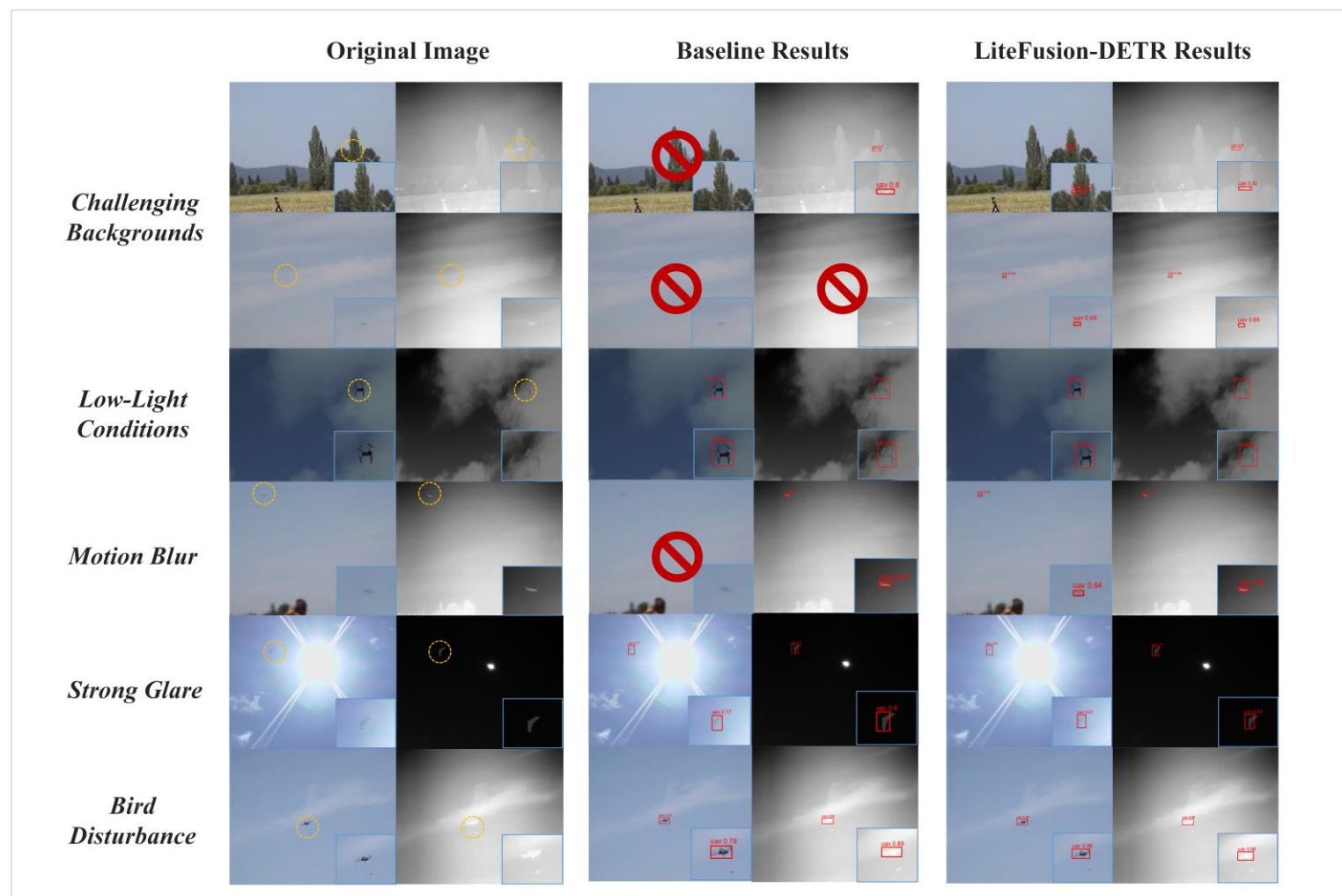
Model	DS	AIFI	PCMGT	DBAF	mAP (%)	Params (M)	FLOPs (G)
RGB only	—	—	—	—	55.78	10.18	24.84
IR only	—	—	—	—	56.56	10.18	24.84
Dual-stream	✓	✓	—	—	61.39	14.13	34.74
Dual-stream	✓	—	—	—	60.70	12.03	33.06
+ PCMGT	✓	—	✓	—	61.81	12.85	34.35
+ DBAF	✓	—	—	✓	62.06	11.77	30.91
Full (ours)	✓	—	✓	✓	63.26	12.64	32.25

mAP (%) by configuration



PCMGT alone **+1.11** mAP · DBAF alone **+1.36** mAP while **reducing** params (−0.26M) and FLOPs (−1.73G) · PCMGT + DBAF **+2.56** mAP over the dual-stream baseline.

Qualitative Results on CUAS-EOIR



How to read

Columns input pair → IR baseline → LiteFusion-DETR

Rows challenging background · low light · motion blur · strong glare · bird disturbance

Observation

The IR baseline misses or mis-localizes the small UAV under glare, blur, and thermal clutter.

LiteFusion-DETR detects the target in **all five conditions**, consistent with the +5.0 mAP50 gain over the IR baseline on CUAS-EOIR.

Conclusion & Future Work

- ① **A simple, strong, efficient baseline.** End-to-end RGB–IR UAV detection with state-of-the-art accuracy at real-time speed.

- ② **PCMGT.** Position-aware gated cross-attention — robust high-level fusion under RGB–IR misalignment.

- ③ **DBAF.** Channel-wise attention — efficient low-level distillation that preserves small-target detail.

- ④ **Verified synergy.** PCMGT + DBAF together yield **+2.56 mAP** over the dual-stream baseline on M3FD.

Future Work

- Region-guided adaptive fusion for severe modality misalignment.
- Extending the framework to heterogeneous sensors.

Thank You

Questions & Discussion

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Zhejiang University · ICPR 2026